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| Inventor(s) | MATSUMOTO; Yuji et al. |


## AIR CONDITIONING SYSTEM AND CONTROL METHOD


### Abstract

An air conditioning system of an embodiment includes an indoor air handling unit controller and an outdoor air handling unit controller. The indoor air handling unit controller is configured to increase and decrease an operating capacity of the indoor air handling unit. The outdoor air handling unit controller is configured to increase and decrease an operating capacity of the outdoor air handling unit.



**Inventors:** **MATSUMOTO; Yuji (Fuji-shi, JP), OKAZAKI; Noriomi (Fuji-shi, JP)**

**Applicant:** **Carrier Japan Corporation** (Tokyo, JP)



**Family ID:** **1000008638960**

**Assignee:** **Carrier Japan Corporation (Tokyo, JP)**

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of International Application No. PCT/JP2023/043166, filed Dec. 1, 2023, which is based upon and claims the benefit of priority from Japanese Patent Application No. 2022-192823, filed on Dec. 1, 2022; the entire contents of which are incorporated herein by reference.

FIELD
[0002] The present invention relates to an air conditioning system and a control method. Priority is claimed on Japanese Patent Application No. 2022-192823 filed on Dec. 1, 2022, the contents of which are incorporated herein by reference.
BACKGROUND
[0003] In conventional technologies, coordinated control of an indoor air handling unit and outdoor air handling unit was aimed at reducing electric energy consumption, and was control of carrying out comparison of a coefficient of performance (COP) or comparison of power consumption (for example, Japanese Patent No. 6414354).

## Description

BRIEF DESCRIPTION OF DRAWINGS
[0004] FIG. **1** is a diagram showing a configuration example of an air conditioning system of an embodiment.
[0005] FIG. **2** is a flowchart showing a specific example of a processing flow of an air conditioning system in a first embodiment.
[0006] FIG. **3** is a flowchart showing a specific example of a processing flow of an air conditioning system in a second embodiment.
DETAILED DESCRIPTION
[0007] Hereinafter, an air conditioning system and a control method according to an embodiment will be described with reference to the drawings.
[0008] An air conditioning system of an embodiment includes an indoor air handling unit controller and an outdoor air handling unit controller. The indoor air handling unit controller is configured to: increase an operating capacity of an indoor air handling unit in a case in which an operating capacity of an outdoor air handling unit is greater than or equal to a first threshold value and the operating capacity of the indoor air handling unit is able to be increased; and decrease the operating capacity of the indoor air handling unit in a case in which the operating capacity of the outdoor air handling unit is less than a second threshold value that is lower than the first threshold value and the operating capacity of the indoor air handling unit can be decreased. The outdoor air handling unit controller is configured to: increase the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is greater than or equal to a third threshold value and the operating capacity of the outdoor air handling unit is able to be increased; and decrease the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is less than a fourth threshold value that is lower than the third threshold value and the operating capacity of the outdoor air handling unit can be decreased.
[0009] FIG. **1** is a diagram showing a configuration example of an air conditioning system **17** of an embodiment. The air conditioning system **17** shown in FIG. **1** is a system that performs air

conditioning for a plurality of spaces (in FIG. **1**, two spaces **23** and **24** provided in a building **14**). The air conditioning system **17** includes a controller **10**, a room-dedicated outdoor unit **12**, and a heat source unit **40**. Also, the building **14** includes an air conditioner controller **18**, two indoor air handling units **21**, an exhaust duct **16**, two variable air volume systems (VAVs) **20**, and two remote controllers **22**. The indoor air handling units **21**, the VAVs **20**, and the remote controllers **22** are each provided in the spaces **23** and **24**.

[0010] The room-dedicated outdoor unit **12** adjusts a temperature of indoor air in the spaces **23** and **24**. The remote controller **22** is used to set starting and stopping of the operation, an airflow, a temperature, and the like. The set contents are output to the controller **10**. The controller **10** controls an operating state of an outdoor air handling unit **31** and the room-dedicated outdoor unit **12** according to the contents set by the remote controller **22**, temperatures and humidities of outdoor air and indoor air, and the like.

[0011] The exhaust duct **16** is connected to an exhaust port (not shown in the drawings) provided in the spaces **23** and **24**. The exhaust duct **16** is a duct for exhausting air in the spaces **23** and **24** to the outside via a building **15**.

[0012] The building **15** includes a ventilation fan **26**, an outdoor air handling unit controller **27**, the outdoor air handling unit **31**, and an air intake port **34**. The ventilation fan **26** exhausts the air from the exhaust duct **16** described above. The outdoor air handling unit controller **27** controls the outdoor air handling unit **31**. The air intake port **34** takes in outdoor air into the outdoor air handling unit **31**.

[0013] The outdoor air handling unit **31** includes an outdoor air handling unit blower **29**, a humidifier **30**, and a water coil **32**. The outdoor air handling unit **31** takes in outdoor air through an air intake duct **28** from the air intake port **34**. The outdoor air handling unit **31** conditions (cools, heats, dehumidifies, or humidifies) the outdoor air taken in. The outdoor air handling unit **31** supplies the conditioned outdoor air from the VAV **20** to the spaces **23** and **24** via a supply air duct **25**. The outdoor air handling unit **31** exhausts air from the spaces **23** and **24** to the outside of the building **13** through the exhaust duct **16** using the ventilation fan **26**.

[0014] The water coil **32** is connected to the heat source unit **40**. The water coil **32** is a heat exchanger that functions as a cooler for the outdoor air. The water coil **32** has a heat transfer tube and a heat transfer fin. In the water coil **32**, heat exchange is performed between outdoor air passing around the heat transfer tube and the heat transfer fin, and a heat medium passing through the heat transfer tube. The humidifier **30** humidifies the outdoor air that has passed through the water coil **32**. The outdoor air handling unit blower **29** is a blower that takes in outdoor air into the outdoor air handling unit **31** and sends it to the supply air duct **25**.

[0015] Various types of sensors are provided in the outdoor air handling unit **31**. Examples of sensors provided in the outdoor air handling unit **31** include an outdoor air temperature sensor and an outdoor air humidity sensor which detect a temperature and a humidity of outdoor air drawn into the outdoor air handling unit **31**. Also, examples of sensors provided in the outdoor air handling unit **31** include a sensor that detects a temperature of the supply air sent to the supply air duct **25**.

[0016] The supply air duct **25** obtains outdoor air by the air blown by the outdoor air handling unit blower **29**. The supply air duct **25** is connected to the outdoor air handling unit **31** and the VAV **20**. The outdoor air handling unit controller **27** is constituted by a calculation device, a storage device, various types of electrical components, and the like. The outdoor air handling unit controller **27** is electrically connected to the controller **10** and the remote controller **22** via a communication line. The outdoor air handling unit controller **27** controls an operating capacity of the outdoor air handling unit **31** according to settings made via the remote controller and a temperature of the outdoor air taken in.

[0017] The room-dedicated outdoor unit **12** generally includes a plurality of outdoor units and a plurality of (two in FIG. **1**) indoor air handling units **21**. The outdoor unit is connected to the indoor air handling unit **21** via a refrigerant communication pipe **19**. Various types of sensors are provided in the outdoor unit and the indoor air handling unit **21**. Examples of sensors provided in the outdoor unit include a suction pressure sensor that detects a pressure of the refrigerant drawn into the compressor, and a discharge pressure sensor that detects a pressure of the refrigerant discharged from the compressor. Examples of sensors provided in the indoor air handling unit **21** include sensors that detect a temperature, a humidity, and a concentration of carbon dioxide of the indoor air drawn into the indoor air handling unit **21**.

[0018] The air conditioner controller **18** is an example of an indoor air handling unit controller. The air conditioner controller **18** is constituted by a calculation device, a storage device, various types of electrical components, and the like. The air conditioner controller **18** is connected to the room-dedicated outdoor unit **12**, the indoor air handling unit **21**, sensors of the indoor air handling unit **21**, and the remote controller **22** via a communication line. The air conditioner controller **18** controls operating capacities of the room-dedicated outdoor unit **12** and the indoor air handling unit **21** according to settings made by the remote controller and an indoor temperature.

[0019] The controller **10** controls the entire air conditioning system **17**. The controller **10** is constituted by a calculation device and a storage device.

[0020] Communication lines L**1**, L**2**, and L**3** are communication lines for performing communication between the air conditioner controller **18** and the outdoor air handling unit controller **27**. The air conditioner controller **18** communicates via the room-dedicated outdoor unit **12**. Also, the outdoor air handling unit controller **27** communicates via the heat source unit **40**. As a mode of performing communication between the air conditioner controller **18** and the outdoor air handling unit controller **27**, there is a first communication mode that uses the communication line L**1**, and a second communication mode that uses the communication lines L**2** and L**3**. The first communication mode is a mode in which communication is performed directly without going through the controller **10**. In a case in which communication is performed in the first communication mode, the communication lines L**2** and L**3** are not necessary. The second communication mode is a mode in which the room-dedicated outdoor unit **12** and the controller **10** are connected by the communication line L**2**, and the heat source unit **40** and the controller **10** are connected by the communication line L**3**. In this way, the second communication mode is a mode in which communication is performed via the controller **10**. In a case in which communication is performed in the second communication mode, the communication line L**1** is not necessary. Note that, as for a communication method, as long as it is sufficient that the air conditioner controller **18** and the outdoor air handling unit controller **27** can communicate with each other, and for example, a general-purpose protocol such as Modbus may be used.

[0021] Based on the above configuration, a first embodiment and a second embodiment will be described.

First Embodiment

[0022] In a first embodiment, the air conditioner controller **18** and the outdoor air handling unit controller **27** are capable of communicating with each other in the first communication mode. The air conditioner controller **18** is configured to be able to acquire an operating capacity of the outdoor air handling unit **31** through communication. The outdoor air handling unit controller **27** is configured to be able to acquire an operating capacity of the indoor air handling unit **21** via communication.

[0023] A first threshold value T**1** for comparison with the operating capacity of the outdoor air handling unit **31** and a second threshold value T**2** smaller than the first threshold value T**1** are preset in the air conditioner controller **18**. The first threshold value T**1** and the second threshold value T**2** are stored in the storage device of the air conditioner controller **18**. The first threshold value T**1** and the second threshold value T**2** are set via, for example, the controller **10**. In a case in which the operating capacity of the outdoor air handling unit **31** is greater than or equal to the first threshold value T**1** and the operating capacity of the indoor air handling unit **21** can be increased, the air conditioner controller **18** increases the operating capacity of the indoor air handling unit **21**.

Also, in a case in which the operating capacity of the outdoor air handling unit **31** is less than the second threshold value T**2** and the operating capacity of the indoor air handling unit **21** can be decreased, the air conditioner controller **18** decreases the operating capacity of the indoor air handling unit **21**. The control for decreasing the operating capacity of the indoor air handling unit **21** also includes control for stopping the operation of the indoor air handling unit **21**.

[0024] A third threshold value T**3** for comparison with the operating capacity of the indoor air handling unit **21** and a fourth threshold value T**4** smaller than the third threshold value T**3** are preset in the outdoor air handling unit controller **27**. The third threshold value T**3** and the fourth threshold value T**4** are stored in the storage device of the outdoor air handling unit controller **27**. The third threshold value T**3** and the fourth threshold value T**4** are set via, for example, the controller **10**. In a case in which the operating capacity of the indoor air handling unit **21** is greater than or equal to the third threshold value T**3** and the operating capacity of the outdoor air handling unit **31** can be increased, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31**. Also, in a case in which the operating capacity of the indoor air handling unit **21** is less than the fourth threshold value T**4** and the operating capacity of the outdoor air handling unit **31** can be decreased, the outdoor air handling unit controller **27** decreases the operating capacity of the outdoor air handling unit **31**. The control for decreasing the operating capacity of the outdoor air handling unit **31** also includes control for stopping the operation of the outdoor air handling unit **31**.

[0025] FIG. **2** is a flowchart showing a processing flow of the air conditioning system **17** in the first embodiment. The air conditioner controller **18** determines whether the operating capacity of the outdoor air handling unit **31** is greater than or equal to the threshold value T**1** (step S**101**). In a case in which the operating capacity of the outdoor air handling unit **31** is greater than or equal to the threshold value T**1** (step S**101**: YES) and in a case in which the operating capacity of the indoor air handling unit **21** can be increased, the air conditioner controller **18** increases the operating capacity of the indoor air handling unit **21** (step S**102**), and then proceeds to step S**105**.

[0026] In a case in which the operating capacity of the outdoor air handling unit **31** is not greater than or equal to the threshold value T**1** (step S**101**: NO), the air conditioner controller **18** determines whether the operating capacity of the outdoor air handling unit **31** is less than the threshold value T**2** (step S**103**). In a case in which the operating capacity of the outdoor air handling unit **31** is less than the threshold value T**2** (step S**103**: YES) and in a case in which the operating capacity of the indoor air handling unit **21** can be decreased, the air conditioner controller **18** decreases the operating capacity of the indoor air handling unit **21** (step S**104**), and then proceeds to step S**105**. In a case in which the operating capacity of the outdoor air handling unit **31** is not less than the threshold value T**2** (step S**103**: NO), the processing proceeds to step S**105**.

[0027] The outdoor air handling unit controller **27** determines whether the operating capacity of the indoor air handling unit **21** is greater than or equal to the threshold value T**3** (step S**105**). In a case in which the operating capacity of the indoor air handling unit **21** is greater than or equal to the threshold value T**3** (step S**105**: YES) and in a case in which the operating capacity of the outdoor air handling unit **31** can be increased, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31** (step S**106**) and ends the processing.

[0028] In a case in which the operating capacity of the indoor air handling unit **21** is not greater than or equal to the threshold value T**3** (step S**105**: NO), the outdoor air handling unit controller **27** determines whether the operating capacity of the indoor air handling unit **21** is less than the threshold value T**4** (step S**107**). In a case in which the operating capacity of the indoor air handling unit **21** is less than the threshold value T**4** (step S**107**: YES) and in a case in which the operating capacity of the outdoor air handling unit **31** can be decreased, the outdoor air handling unit controller **27** decreases the operating capacity of the outdoor air handling unit **31** (step S**108**) and ends the processing. In a case in which the operating capacity of the indoor air handling unit **21** is not less than the threshold value T**4** (step S**107**: NO), the processing ends.

[0029] The processing shown in FIG. **2** may be configured as loop processing that does not end at step S**106** or step S**108**, but returns to step S**101** after step S**106** or step S**108** is completed. Alternatively, this processing may be repeatedly performed at regular intervals (for example, every one minute). This processing may also be performed in response to an instruction from an operator of the air conditioning system **17**.
[0030] A flow of the basic processing in the first embodiment is as described above. As shown in the flowchart of FIG. **2**, a purpose of the processing in the present embodiment is to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. In a case in which it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**, processing may be added to or modified in the processing shown in the flowchart of FIG. **2**.
[0031] Specifically, a processing example in a case in which the threshold values T**1** and T**3** are set to an operating capacity of 80%, and the threshold values T**2** and T**4** are set to an operating capacity of 70% will be described. First, in a case in which the operating capacity of the indoor air handling unit **21** reaches 80% or more, if set dehumidification performance is obtained, the air conditioner controller **18** increases an airflow of the indoor air handling unit **21**, and the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit. In a case in which the operating capacity of the indoor air handling unit **21** is less than the threshold value T**1** and greater than or equal to the threshold value T**3**, the outdoor air handling unit controller **27** does not perform airflow correction control, but instead controls the operating capacity to be equalized with that of the indoor air handling unit **21**. In a case in which the operating capacity of the outdoor air handling unit **31** is less than 70%, the air conditioner controller **18** gradually reduces the airflow correction and shifts to normal control. Here, the normal control refers to control of the room-dedicated outdoor unit **12** and the indoor air handling unit **21** that is performed according to settings made by the remote controller or an indoor temperature.
[0032] Note that, for example, in a case in which the operating capacity of the indoor air handling unit **21** reaches 80% or more and in a case in which the operating capacity of the outdoor air handling unit **31** is assumed to be 100%, the operating capacity of the outdoor air handling unit **31** cannot be increased. However, in a case in which the ventilation airflow is satisfactory, the outdoor air handling unit controller **27** reduces the load on the outdoor air handling unit **31** (reduces the airflow, changes the set temperature of the heat source unit, or reduces a water flow rate). Therefore, since the load on the outdoor air handling unit **31** can be reduced while the load on the indoor air handling unit **21** increases, it is possible to suppress the load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**.
[0033] In a case in which controlled in this way, it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. Therefore, since it is possible to avoid situations in which components of equipment operating under a high load are more likely to fail or a maintenance cycle of the equipment operating under a high load becomes shorter, running costs of the air conditioning system can be reduced.
Second Embodiment
[0034] In the second embodiment, the air conditioner controller **18** and the outdoor air handling unit controller **27** can communicate with each other in the first communication mode or the second communication mode. The air conditioner controller **18** is configured to be able to acquire an operating capacity of the outdoor air handling unit **31** via communication. The outdoor air handling unit controller **27** is configured to be able to acquire an operating capacity of the indoor air handling unit **21** via communication.
[0035] A predetermined value A (>0) for comparing a difference between the operating capacity of the outdoor air handling unit **31** and the operating capacity of the indoor air handling unit **21** is preset in the air conditioner controller **18**. The predetermined value A is stored in the storage device of the air conditioner controller **18**. Similarly, the predetermined value A is preset in the outdoor air

handling unit controller **27**. The predetermined value A is stored in the storage device of the outdoor air handling unit controller **27**. The predetermined value A is set via, for example, the controller **10**. In a case in which a difference between an operating capacity X of the outdoor air handling unit **31** and an operating capacity Y of the indoor air handling unit **21** becomes greater than or equal to the predetermined value A ($X-Y>A$) and the operating capacity of the indoor air handling unit **21** can be increased, the air conditioner controller **18** increases the operating capacity of the indoor air handling unit **21**. Also, in a case in which the difference between the operating capacity of the outdoor air handling unit **31** and the operating capacity of the indoor air handling unit **21** becomes less than the predetermined value ($X-Y<A$) and the operating capacity of the indoor air handling unit **21** can be decreased, the air conditioner controller **18** decreases the operating capacity of the indoor air handling unit **21**. The control for decreasing the operating capacity of the indoor air handling unit **21** also includes control for stopping the operation of the indoor air handling unit **21**.

[0036] In a case in which a difference between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** becomes greater than or equal to the predetermined value A ($Y-X>A$) and the operating capacity of the outdoor air handling unit **31** can be increased, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31**. Also, in a case in which the difference between the operating capacity of the indoor air handling unit **21** and the operating capacity of the outdoor air handling unit **31** becomes less than the predetermined value ($Y-X<A$) and the operating capacity of the outdoor air handling unit **31** can be decreased, the outdoor air handling unit controller **27** decreases the operating capacity of the outdoor air handling unit **31**. The control for decreasing the operating capacity of the outdoor air handling unit **31** also includes control for stopping the operation of the outdoor air handling unit **31**.

[0037] FIG. **3** is a flowchart showing a processing flow of the air conditioning system **17** in the second embodiment. In the flowchart of FIG. **3**, a difference ($X-Y$) between an operating capacity X of the outdoor air handling unit **31** and an operating capacity Y of the indoor air handling unit **21** is defined as Z. A difference ($Y-X$) between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is defined as W.

[0038] The air conditioner controller **18** determines whether the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** is greater than or equal to the predetermined value A (step S**201**). In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** is greater than or equal to the predetermined value A (step S**201**: YES) and in a case in which the operating capacity of the indoor air handling unit **21** can be increased, the air conditioner controller **18** increases the operating capacity of the indoor air handling unit **21** (step S**202**), and then proceeds to step S**205**.

[0039] In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** is not greater than or equal to the predetermined value A (step S**201**: NO), the air conditioner controller **18** determines whether the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step S**203**). In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step S**203**: YES), and in a case in which the operating capacity of the indoor air handling unit **21** can be decreased, the air conditioner controller **18** decreases the operating capacity of the indoor air handling unit **21** (step S**204**), and then proceeds to step S**205**. In a case in which the difference Z between the operating capacity X of the outdoor air handling unit **31** and the operating capacity Y of the indoor air handling unit **21** has not become less than the predetermined value A from a state greater than or equal to the predetermined value A (step S**203**: NO), the processing proceeds to step S**205**.

[0040] The outdoor air handling unit controller **27** determines whether a difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is greater than or equal to the predetermined value A (step S**205**). In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is greater than or equal to the predetermined value A (step S**205**: YES) and in a case in which the operating capacity of the outdoor air handling unit **31** can be increased, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31** (step S**206**) and ends the processing.

[0041] In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is not greater than or equal to the predetermined value A (step S**205**: NO), the outdoor air handling unit controller **27** determines whether the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step S**207**). In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** has become less than the predetermined value A from a state greater than or equal to the predetermined value A (step S**207**: YES), and in a case in which the operating capacity of the outdoor air handling unit **31** can be decreased, the outdoor air handling unit controller **27** decreases the operating capacity of the outdoor air handling unit **31** (step S**208**) and ends the processing. In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** has not become less than the predetermined value A from a state greater than or equal to the predetermined value A (step S**207**: NO), the processing ends.

[0042] The above-described processing shown in FIG. **3** may be performed at regular intervals (for example, every one minute) or in response to an instruction from an operator of the air conditioning system **17**.

[0043] A flow of the basic processing in the second embodiment is as described above.

[0044] As shown in the flowchart of FIG. **3**, a purpose of the processing in the present embodiment is to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. In a case in which it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**, processing may be added to or modified in the processing shown in the flowchart of FIG. **3**.

[0045] Specifically, a processing example in a case in which the predetermined value A is set to 20% will be described. In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** is 20% or more, this is assumed to be an operating capacity at which dehumidification performance of the outdoor air handling unit **31** can be obtained. In this case, the outdoor air handling unit controller **27** increases the operating capacity of the outdoor air handling unit **31** by increasing an airflow of the outdoor air handling unit **31** or lowering a blowing temperature. As a result, since the load is equalized, it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. In a case in which the difference W between the operating capacity Y of the indoor air handling unit **21** and the operating capacity X of the outdoor air handling unit **31** becomes less than 20% from a state of 20% or more, the air conditioner controller **18** stops the operation of the indoor air handling unit **21**, and the outdoor air handling unit controller **27** performs normal control using only the outdoor air handling unit **31**. The normal control here refers to control by the outdoor air handling unit controller **27** that is performed according to settings made by the remote controller or a temperature of outdoor air taken in. As equipment having a higher load between the outdoor air handling unit **31** or the indoor air handling

unit **21** is stopped in this way, an operation time of the equipment with a higher load is reduced.
[0046] In this way, it is possible to suppress a load becoming biased toward either the outdoor air handling unit **31** or the indoor air handling unit **21**. Therefore, since it is possible to avoid situations in which components of equipment operating under a high load are more likely to fail or a maintenance cycle of the equipment operating under a high load becomes shorter, running costs of the air conditioning system can be reduced.
[0047] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

## Claims

**1**. An air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising: an indoor air handling unit controller configured to increase an operating capacity of the indoor air handling unit in a case in which an operating capacity of the outdoor air handling unit is greater than or equal to a first threshold value and the operating capacity of the indoor air handling unit is able to be increased, and decrease the operating capacity of the indoor air handling unit in a case in which the operating capacity of the outdoor air handling unit is less than a second threshold value that is lower than the first threshold value and the operating capacity of the indoor air handling unit is able to be decreased; and an outdoor air handling unit controller configured to increase the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is greater than or equal to a third threshold value and the operating capacity of the outdoor air handling unit is able to be increased, and decrease the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is less than a fourth threshold value that is lower than the third threshold value and the operating capacity of the outdoor air handling unit is able to be decreased.
**2**. An air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising: an indoor air handling unit controller configured to increase an operating capacity of the indoor air handling unit in a case in which a difference between an operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the indoor air handling unit is able to be increased, and decrease the operating capacity of the indoor air handling unit in a case in which the difference between the operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes less than the predetermined value and the operating capacity of the indoor air handling unit is able to be decreased; and an outdoor air handling unit controller configured to increase the operating capacity of the outdoor air handling unit in a case in which a difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the outdoor air handling unit is able to be increased, and decrease the operating capacity of the outdoor air handling unit in a case in which the difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes less than the predetermined value and the operating capacity of the outdoor air handling unit is able to be decreased.
**3**. A control method of an air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising: increasing an operating capacity of the indoor air handling unit in a case in which an operating capacity of the outdoor air handling unit is greater than or

equal to a first threshold value and the operating capacity of the indoor air handling unit is able to be increased, and decreasing the operating capacity of the indoor air handling unit in a case in which the operating capacity of the outdoor air handling unit is less than a second threshold value that is lower than the first threshold value and the operating capacity of the indoor air handling unit is able to be decreased; and increasing the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is greater than or equal to a third threshold value and the operating capacity of the outdoor air handling unit is able to be increased, and decreasing the operating capacity of the outdoor air handling unit in a case in which the operating capacity of the indoor air handling unit is less than a fourth threshold value that is lower than the third threshold value and the operating capacity of the outdoor air handling unit is able to be decreased.

**4**. A control method of an air conditioning system including an indoor air handling unit and an outdoor air handling unit, comprising: increasing an operating capacity of the indoor air handling unit in a case in which a difference between an operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the indoor air handling unit is able to be increased, and decreasing the operating capacity of the indoor air handling unit in a case in which the difference between the operating capacity of the outdoor air handling unit and the operating capacity of the indoor air handling unit becomes less than the predetermined value and the operating capacity of the indoor air handling unit is able to be decreased; and increasing the operating capacity of the outdoor air handling unit in a case in which a difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes greater than or equal to a predetermined value and the operating capacity of the outdoor air handling unit is able to be increased, and decreasing the operating capacity of the outdoor air handling unit in a case in which the difference between the operating capacity of the indoor air handling unit and the operating capacity of the outdoor air handling unit becomes less than the predetermined value and the operating capacity of the outdoor air handling unit is able to be decreased.